Lucid: End-to-End Data Ingestion & Agent Delivery

Five-Stage Pipeline: Free-Tier External Ingestion, Normalization, Agent Synthesis & Math Allocation

STAGE 1: EXTERNAL FREE-TIER PUBLIC DATA SOURCES

5 VERIFIED AUTHORITIES

NSE & BSE Archives
2,639 Master Equities
ISIN Registry & Tickers
Weekly Batch EOD
Fallback: PostgreSQL Cache

Yahoo Finance API
1-Year Daily OHLCV
P/E, P/B, ROE, D/E
15-Min Lag / Post-Close
Fallback: Stored 252 Bars

Reserve Bank (RBI)
Repo 6.50% • CPI 5.10%
10Y G-Sec: 7.04%
Bi-Monthly MPC Bulletins
Fallback: Seeded 6.50%

NSE Stream & News
India VIX & Nifty 50
30-Day News Headlines
15-Min Public Index Lag
Fallback: 20-DMA Regime

Google Gemini API
gemini-2.5-flash
Qualitative Reasoning
1.5s-2.5s Roundtrip Quota
Fallback: Rule Engine v2

HTTPS REST & CSV

STAGE 2: DATA NORMALIZATION, FUNNEL & RELATIONAL PERSISTENCE

POSTGRESQL 16.4

1. Dual-Listing & ISIN Resolver
- Ticker Mapping: NSE:RELIANCE ↔ BSE:500325
- ISIN Validation: INE002A01018 verification
- Cleanse symbol aliases & corporate splits

Saved: securities & instrument_listings

2. 3-Tier Liquidity Funnel
- Master Universe: 2,639 listed equities
- **Liquidity Gate: ADTV ≥ ₹5 Lakhs, 200+ days**
- **Liquid Universe Output: Exactly 504 Stocks**

Filters penny stocks & illiquid securities

3. Factor Winsorization & Z-Scores
- Outlier Clipping: [5th, 95th Percentile]
- Quality Z_Q: Sector-relative ROE
- Value Z_V: P/E, P/B; Momentum Z_M (6M)

Saved: fundamental_metrics, market_prices

Normalized Context Ingestion

STAGE 3: STRUCTURED AGENT PAYLOAD & GRAPH STATE ASSEMBLY

PortfolioAgentState (35 Fields)

Investor Mandate
- Capital: ≥ ₹10,000
- 5-Q Risk Score (5-20), Horizon, Sector Bans

Macroeconomic Context
- RBI Repo: 6.50% • CPI Inflation: 5.10%
- India VIX Regime: Calm, Moderate, Volatile

Pre-Screened Universe
- 504 Verified Liquid Stocks
- Winsorized Factor Z-Scores & Valuations

Dispatched to LangGraph Multi-Agent Team

STAGE 4: LANGGRAPH MULTI-AGENT SEQUENTIAL RESEARCH & ADVERSARIAL CRITIC

QUALITATIVE REASONING

1. Chained Specialist Research Analysts (Enrich Shared Graph State Sequentially)

Macro Analyst (N3) → Repo, CPI → Cash Buffer 2.5–5%

Market Analyst (N4) → VIX, 200-DMA → Beta & Vol Cap

Fundamental Analyst (N5) → ROE ≥15%, D/E ≤1.0, P/E Screening

News & Filings (N6) → 30d News, Filings, Pledges <5%

2. Strategist Allocation & Adversarial Critic Feedback Loop

Lead Strategist Node (N7)
Selects K ∈ [10..16] Candidate Stocks
Core • Growth • Defensive

→

Adversarial Critic Gate (N8)
Audits Mandate, Caps (≤25%) & Ethics
Rework Loop (≤2) → Swaps Tickers

→

Vetted Candidates
10–16 Critic-Approved Equities
Passed to Deterministic Math

Architectural Boundary: Qualitative AI → Zero-LLM Math

STAGE 5: ZERO-LLM DETERMINISTIC FINANCIAL MATH & WHOLE-SHARE ALLOCATION

NUMERICAL VERIFICATION

1. SciPy SLSQP Solver (N10)
- Max Sharpe: max [$w^T\mu$ - Rf] / σ_p
- Hard Bounds: $0.05 \le w_i \le 0.25$
- Sector Cap: $\sum w_s \le 0.40$

Solves continuous target weights w_i^*

2. Whole-Share Rounding (N10B)
- Integer Lots: $Q_i = \lfloor (w_i^* \cdot V_{tot}) / P_i \rfloor$
- Greedy Remainder Pass on Cash
- **Guaranteed Cash Reserve: $C_{res} \ge$ ₹0**

No fractional shares; 100% executable

3. Invariant Audit & Levies (N11)
- Verification: Post-Rounding Caps
- Statutory Indian Levies (Oct 2024):
  STT (0.10%) • Stamp • Exchange • GST

Node 11 Audit Gate: Zero violations

Fiduciary Output Delivery (Nodes 12 & 13):
- Two-Level Explainability Modal (Level-1 Plain-English Thesis + Level-2 Official NSE/BSE Announcement Proofs)

Atomic ACID Commit
6 Tables (Versions, Decisions, Audit)

Pipeline Summary: Raw Public Data → Validated Liquid Pool (504) → Multi-Agent Vetting (10-16) → SLSQP Solved Integer Portfolio